USN | | | | | | | | | | | [22CSE529]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

ADVANCED COMPUTER NETWORKS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A	**5 x2=10**		
1.	**a.**	What is the primary function of the Domain Name System (DNS) in the application layer?	L2	CO1	**02**
	b.	Explain the purpose of TCP flow control.	L2	CO2	**02**
	c.	How VLSM is useful in IP addressing?	L2	CO3	**02**
	d.	What is the purpose of the Spanning Tree Protocol in switched networks?	L2	CO4	**02**
	e.	In the context of Wi-Fi networks, what is the significance of radio frequency?	L2	CO5	**02**
		PART – B	**6 x5=30**		
2.	**a.**	Describe the layered operation in networking and explain the functions of the Application Layer.	L2	CO1	**05**
		OR			
	b.	Explain the working of the HTTP protocol and its role in the World Wide Web.	L2	CO1	**05**
3.	**a.**	Compare and contrast TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).	L2	CO2	**05**
		OR			
	b.	Describe the TCP connection establishment process and explain the purpose of the three-way handshake.	L2	CO2	**05**
4.	**a.**	Discuss the limitations of IPv4 and the key features of IPv6.	L2	CO3	**05**
		OR			
	b.	Explain the concept of CIDR (Classless Inter-Domain Routing) and how it improves upon the earlier classful addressing system.	L2	CO3	**05**
5.	**a.**	Describe the CSMA/CD protocol used in Ethernet networks and explain how it handles collisions.	L2	CO4	**05**
		OR			
	b.	Explain the concept of VLANs (Virtual Local Area Networks) and discuss their benefits in network design.	L2	CO4	**05**

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6.	a.	Describe the key components and operation of 802.11 (Wi-Fi) networks.	L2	CO5	**05**
		OR			
	b.	Illustrate the purpose and functioning of the Real-time Transport Protocol (RTP) in multimedia communications.	L2	CO5	**05**
7.	a.	Explain TCP congestion control mechanisms and their importance in the network performance.	L2	CO2	**05**
		OR			
	b.	Compare and contrast distance vector routing protocols and link-state routing protocols.	L2	CO3	**05**
		PART – C	**6 X 10=60**		
8.	a.	Analyze the hierarchical structure of DNS and explain how a recursive DNS query works. Discuss two potential DNS vulnerabilities and their mitigations.	L4	CO1	**10**
		OR			
	b.	Compare and contrast the SMTP and FTP protocols in termsof their functionality, typical use cases, and connection management. How does each protocol handle data transfer?	L3	CO1	**10**
9.	a.	Explain how TCP congestion control mechanisms (slow start, congestion avoidance, and fast recovery) work together to manage network congestion. Discuss the benefits of implementing Random Early Detection (RED) in a content delivery network (CDN).	L2	CO2	**10**
		OR			
	b.	Explain the concept of TCP flow control and how it differs from congestion control. Describe the sliding window protocol and its role in TCP flow control.	L2	CO2	**10**
10.	a.	Explain the concepts of VLSM with an example network.	L2	CO3	**10**
		OR			
	b.	Describe the process of subnetting in IPv4 networks. How does CIDR (Classless Inter-Domain Routing) improve upon the original classful addressing system? Provide an example of how CIDR notation is used.	L2	CO3	**10**

11.	**a.**	Design a WAN solution for a company with a headquarter and five branch offices using Frame Relay and ATM. Address topology, QoS for voice and video traffic, security considerations, and inter-VLAN routing. Justify your design choices.	L3	CO4	**10**
		OR			
	b.	Explain the concept of VLANs (Virtual Local Area Networks)and Inter VLAN with their benefits in network design.	L2	CO4	**10**
12.	**a.**	Explain how CSMA/CA works in this context. Discuss two security challenges in Wi-Fi networks and their solutions.	L2	CO5	**10**
		OR			
	b.	Compare and contrast Wi-Fi 6 (802.11ax) with previous Wi- Fi standards. What are the key features that improve performance and efficiency in Wi-Fi 6? How does Wi-Fi 6 address issues in high-density deployments?	L2	CO5	**10**
13.	**a.**	Explain the concept of Weighted Fair Queuing (WFQ) in network traffic management. How does it differ from simpleFirst-In-First-Out (FIFO) queuing? Discuss the benefits and potential drawbacks of implementing WFQ in a large enterprise network.	L2	CO2	**10**
		OR			
	b.	Describe the key components and operation of Frame Relay networks. Explain the concepts of CIR (Committed Information Rate) and burst traffic in the context of Frame Relay.	L2	CO4	**10**